

DLA: Autosomal recessive disorders

GNET DLA_Before Mendelian Inheritance: AD and AR

Three parts:

- A. Reading assignment
- B. Autosomal dominant
- C. **Autosomal recessive**

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Inheritance: AD and AR – *DLAs*

SOM.MK.III.BPM1.1.FTM.3.GNET.0020	Describe basic definitions of human genetics including: locus, gene, allele, genotype, and zygosity
SOM.MK.III.BPM1.1.FTM.3.GNET.0021	From a description of a pedigree, identify autosomal dominant (AD) and autosomal recessive (AR) inheritance; Compare and contrast Mendelian and non-Mendelian genetics
SOM.MK.III.BPM1.1.FTM.3.GNET.0022	Calculate recurrence risk of autosomal dominant and autosomal recessive disorders within families
SOM.MK.III.BPM1.1.FTM.3.GNET.0023	Analyze family pedigree to determine the risk that immediate family members might be carriers of disease alleles (AR or X-linked) and when to use the 2/3 rule
SOM.MK.III.BPM1.1.FTM.3.GNET.0024	Describe and differentiate between pseudoautosomal dominance, co-dominance, and incomplete dominance; give examples
SOM.MK.III.BPM1.1.FTM.3.GNET.0025	Describe co-dominance, giving examples
SOM.MK.III.BPM1.1.FTM.3.GNET.0026	Describe incomplete dominance, giving examples

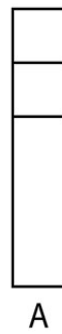
Inheritance: AD and AR - *Flipped classroom lecture*

SOM.MK.III.BPM1.1.FTM.3.GNET.0027	Interpret a family pedigree incorporating standard genetic symbols. Interpret a pedigree and identify features, disorders, genotype, carrier status, recurrence risk, and genetic principles
SOM.MK.III.BPM1.1.FTM.3.GNET.0028	Demonstrate how disease liability is transmitted for autosomal dominant and autosomal recessive disorders, beginning from the first "de novo" occurrence
SOM.MK.III.BPM1.1.FTM.3.GNET.0029	Propose how the basic risk calculations are performed for the different categories of genetic transmission: AR and AD
SOM.MK.III.BPM1.1.FTM.3.GNET.0030	Identify obligate carriers in a pedigree: AR, or AD with reduced penetrance
SOM.MK.III.BPM1.1.FTM.3.GNET.0031	Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following autosomal dominant disorders: Huntington disease, myotonic dystrophy, familial hypercholesterolemia, osteogenesis imperfecta, Marfan syndrome, Achondroplasia, NF, acute intermittent porphyria)
SOM.MK.III.BPM1.1.FTM.3.GNET.0032	Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following following autosomal recessive disorders: CF, PKU, Tay-Sachs disease, Sickle cell disease, thalassemia, hemochromatosis, homocystineuria, SCID due to ADA deficiency, alkaptonuria
SOM.MK.III.BPM1.1.FTM.3.GNET.0033	Identify pseudodominant inheritance and indicate factors that result in this inheritance pattern. Be able to identify this concept from clinical scenarios and pedigrees
SOM.MK.III.BPM1.1.FTM.3.GNET.0034	Compare and contrast locus heterogeneity and allelic heterogeneity; give examples
SOM.MK.III.BPM1.1.FTM.3.GNET.0035	Explain how allelic heterogeneity allows the formation of a compound heterozygote in autosomal recessive disorders
SOM.MK.III.BPM1.1.FTM.3.GNET.0036	Define the terms and give examples for: haploinsufficiency, dominant negative effect, gain of function mutations, loss of function mutations

In this brief video (and throughout the course), examples of autosomal recessive disorders will be presented to illustrate genetic principles

- Autosomal recessive:
 - Parents of an affected individual are carriers (typically) and are NOT affected
 - Unaffected siblings of a patient have a $2/3$ risk of being a carrier
 - When both parents are carriers, the risk to future offspring is 25% ($1/4$)
- Draw Punnett squares for recurrence risk calculations
- Generally, loss-of function mutations
- Many of the disorders have **newborn screening** for early identification and treatment critical to prevent complications and progression
 - Cystic fibrosis, PKU, galactosemia

Autosomal recessive inheritance



A

Dominant allele



a

Recessive allele

Parents of affected children: Carriers

Children of affected parent: Carriers

Calculate:

Risk of disorder in future offspring of II-2 and II-3: **25% (1/4)**

Carrier risk of III-1 and III-2: **67% or 2/3**

**Carrier/
Heterozygote**



A



A



A



a

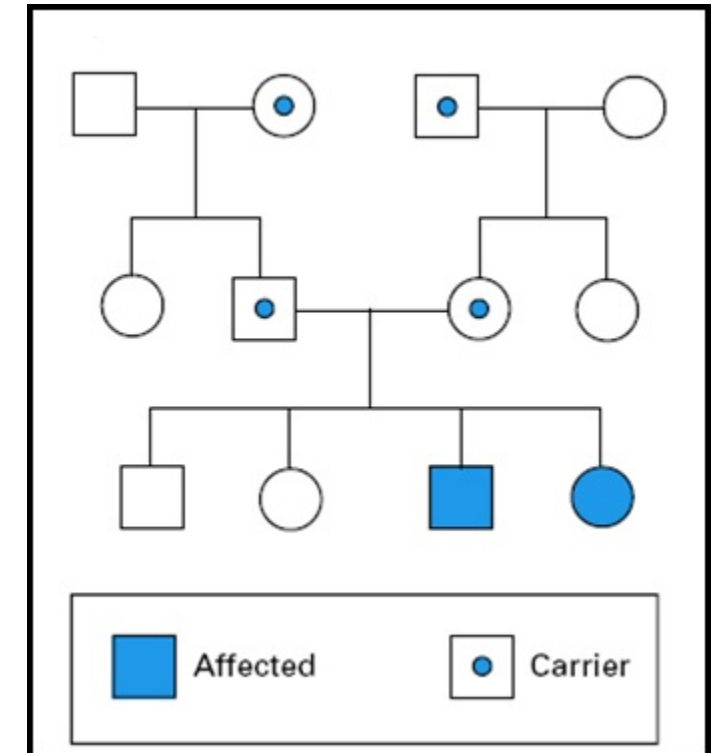


a



a

Genotype	Homozygous	Heterozygous	Homozygous
Phenotype	Unaffected	Unaffected	Affected



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Figure 3.2 Genotype and phenotype associated with a gene locus with dominant allele A and recessive allele a.

Autosomal Recessive disorders

We expect you to remember these disorders, their gene names and their mode of transmission – each will be expanded upon as we progress through this term, and on.

You will see all of these in detail as we progress

- Cystic fibrosis
 - *CFTR* gene, chloride channel, highly penetrant
- Sickle cell anemia (sickle cell disease)
 - Structural variant of the beta-globin protein, which is part of hemoglobin
 - Single mutation causes the disease, causes Glu6Val in protein
 - When homozygous the protein tends to polymerize, distorts RBC
- Phenylketonuria (*PAH* gene)
 - Inability to degrade phenylalanine (an amino acid)
 - Phenylalanine builds up, causes brain damage
- Tay-Sachs disease (Hexosaminidase A deficiency *HEXA* gene)
- Hemochromatosis (delayed age of onset, and incomplete penetrance)
 - Tendency to accumulate toxic amounts of iron
- Homocystinuria (build up of homocysteine)
- Galactosemia (inability to degrade milk sugar; galactose)
- Alpha 1-antitrypsin deficiency: adult-onset emphysema, and liver damage in children
- SCID due to adenosine deaminase deficiency (*ADA* gene)
 - Severe combined immune deficiency due to build up of adenosine
- Most **enzyme deficiencies are AR**
 - Beware those that are X-linked, and AIP (Acute intermittent porphyria which is AD)

Galactosemia

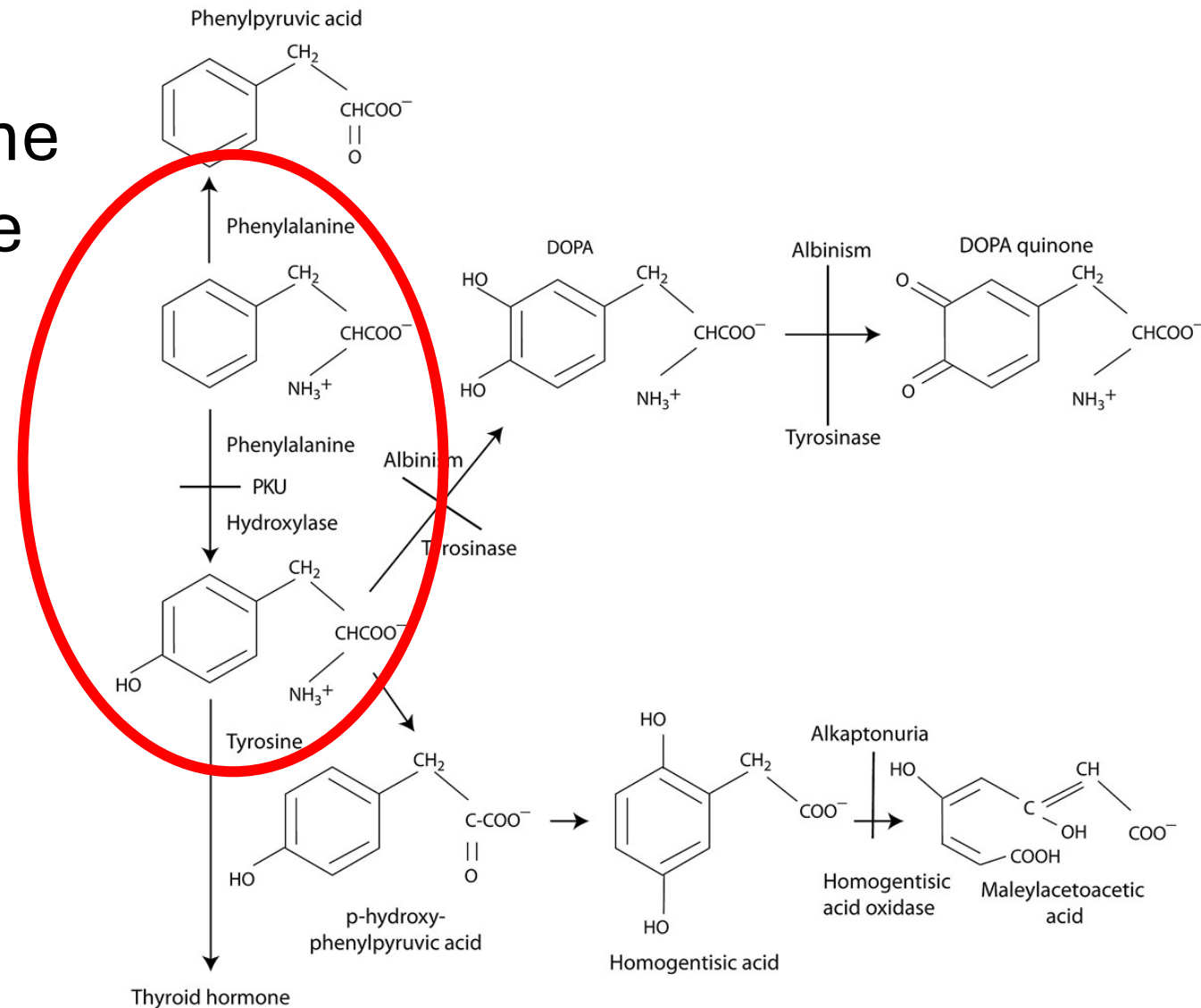
- Homozygous loss of function in an enzyme required for **metabolism of galactose**
- Lactose (milk sugar) is a dimer of Glucose and Galactose
- Liver damage, cataracts and intellectual disability
- Newborn screening
- Treatment: Restriction of milk sugar (galactose) in the diet
- Differentiate from lactose intolerance (defect in intestinal lactose digestion)

Phenylketonuria: PKU

- Inheritance of two loss of function alleles of the enzyme **phenylalanine hydroxylase** (*PAH* gene)
- Allelic heterogeneity
 - Different types of mutation of the PAH gene can result in PKU
 - Compound heterozygote: In a patient with PKU, genetic analysis shows 2 different mutations of the PAH gene
- Buildup of Phe is toxic, causes irreversible brain damage
- Newborn screening in most developed countries
- Management is low Phe diet

PKU – inability to convert phenylalanine to tyrosine (both are amino acids)

Treatment: dietary restriction of phenylalanine



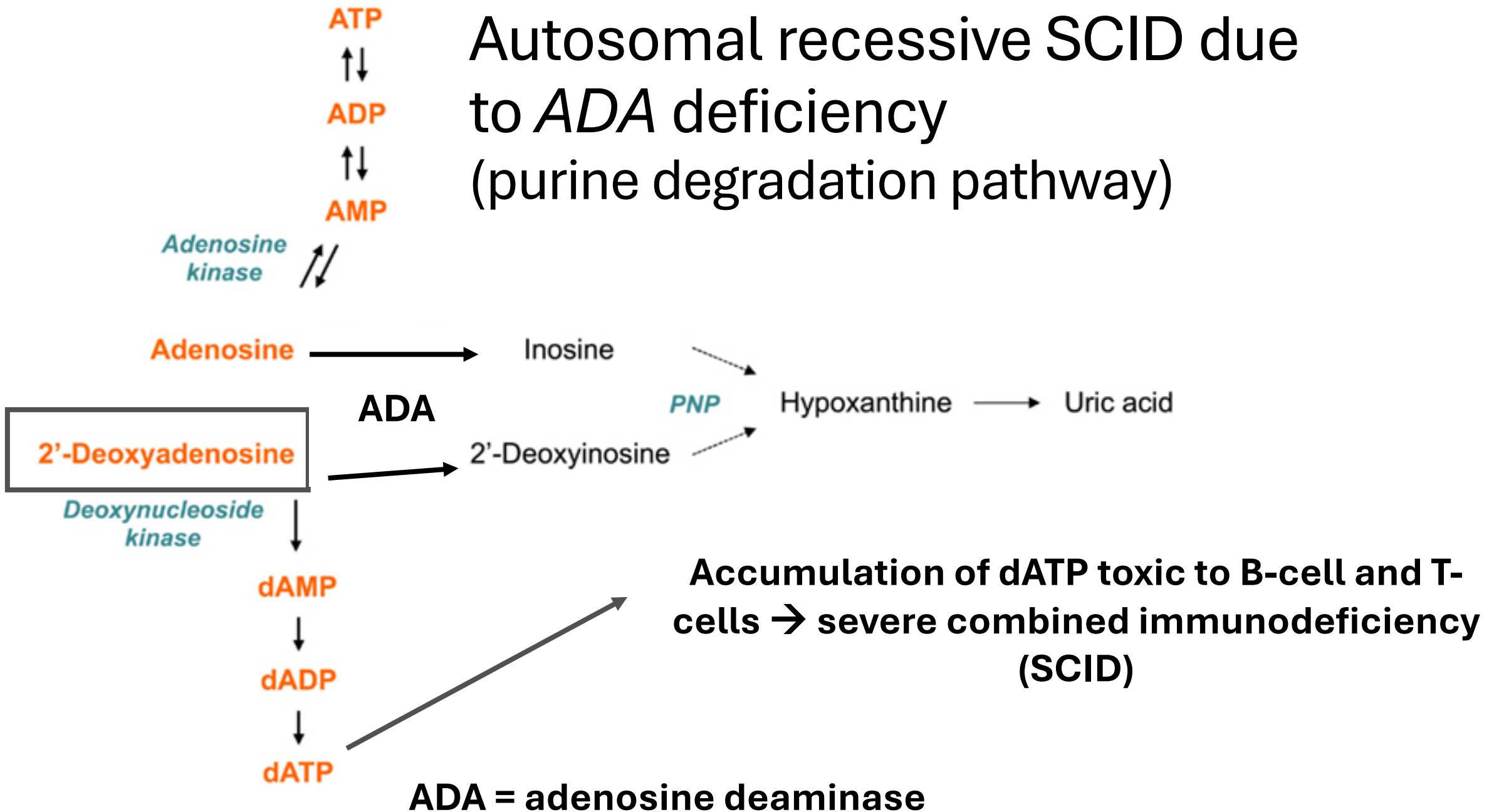
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Figure 3.5 Metabolic pathways involving tyrosine. Phenylalanine is converted to tyrosine by phenylalanine hydroxylase, the enzyme blocked in phenylketonuria. Lack of homogentisic acid oxidase leads to alkaptonuria and deficiency of tyrosinase to albinism.

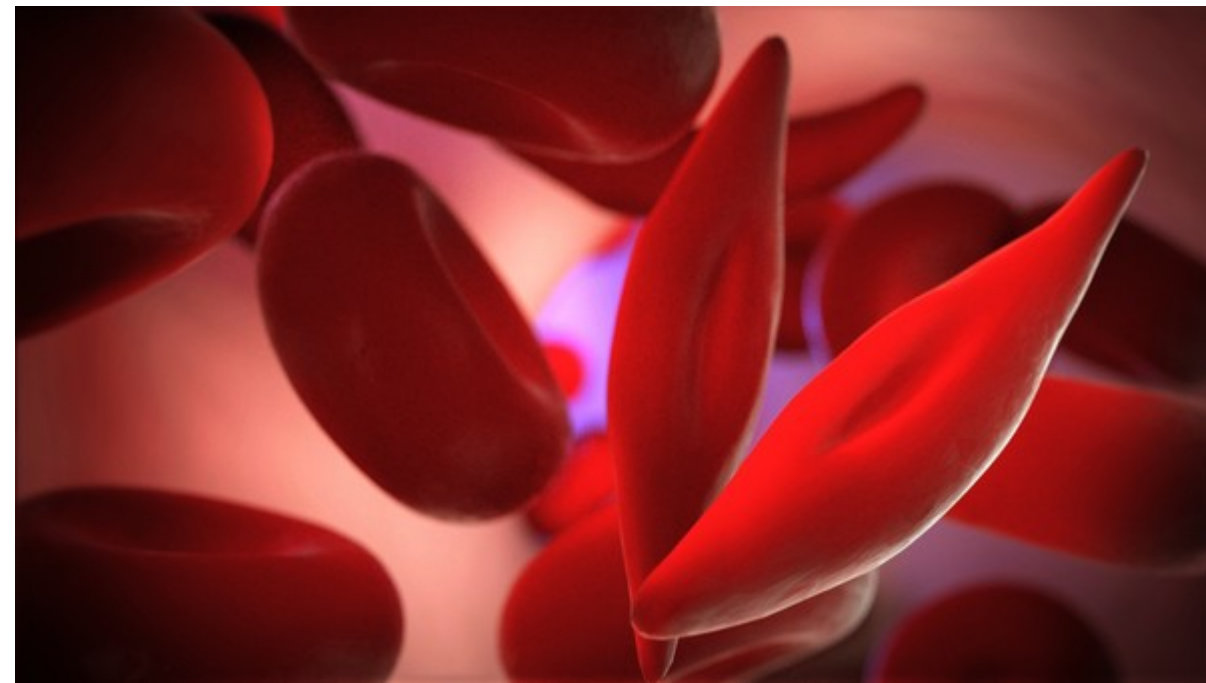
Homocystinuria

- Individuals affected with homocystinuria may develop a body shape resembling Marfan syndrome
- High levels of homocysteine interacts with connective tissue and also cause neurological manifestations
- Autosomal recessive
- Inborn error of metabolism due to an enzyme deficiency

Autosomal recessive SCID due to *ADA* deficiency (purine degradation pathway)



Sickle cell disease or sickle cell anemia



Credit: [U.S. National Library of Medicine](#), [Use of illustrations and other content](#)

- Point mutation in the gene that encodes beta-globin (in hemoglobin)
- Glu6Val (Glutamate → Valine) change in the encoded protein
- Causes hemoglobin to polymerize → Alters RBC shape
- “Gain of function”
- Causes painful vaso-occlusive crisis
- HbS can still bind and release oxygen

Sickle cell trait: Heterozygous carrier

- At risk of having a sickling crisis under extreme conditions
- For example: high altitude + vigorous exercise
- Hence, a phenotype can sometimes be uncovered in the otherwise “unaffected heterozygote” carrier

Sickle cell disease vs sickle cell trait

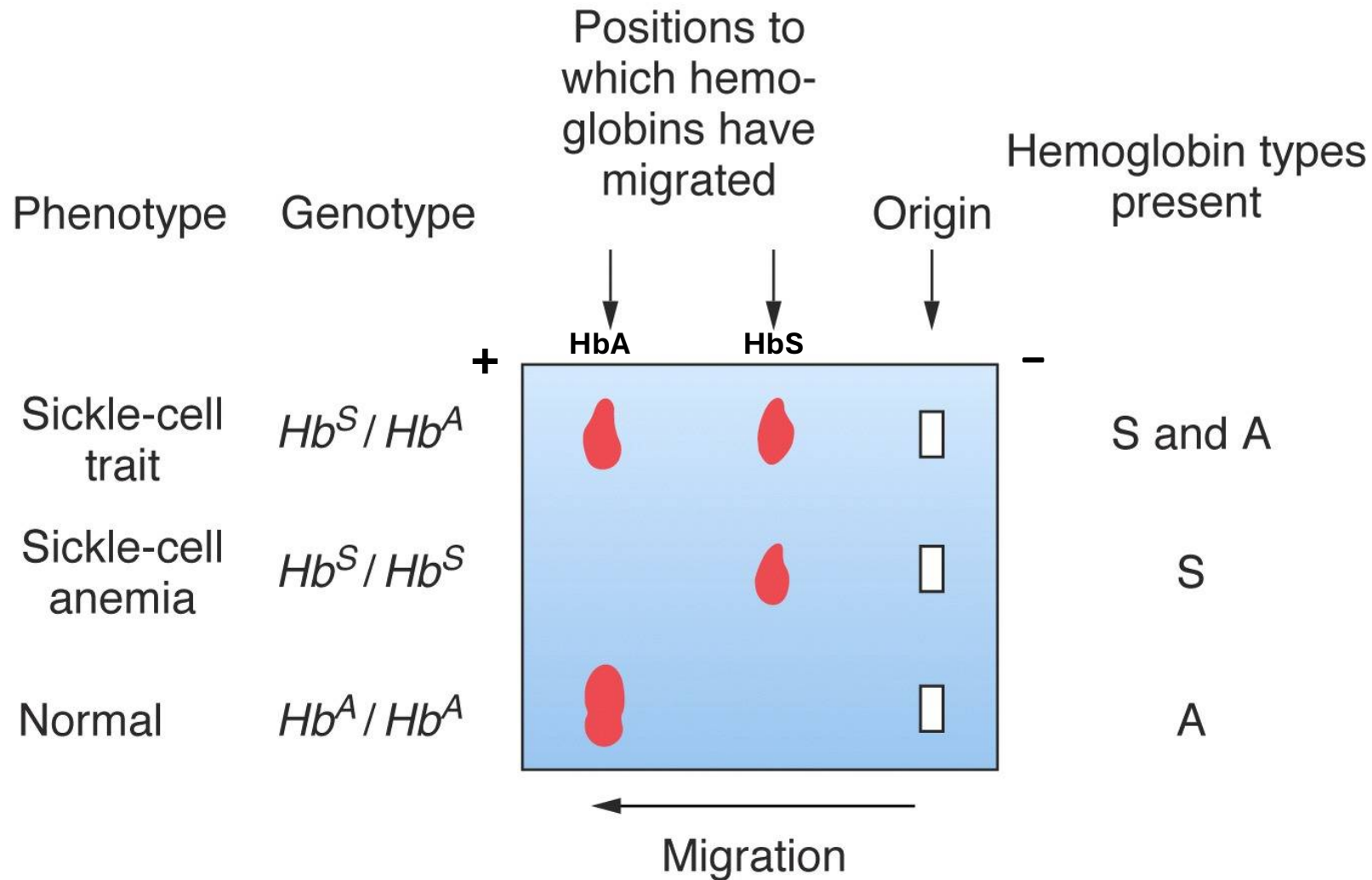
Sickle cell disease

- Genotype SS
- **Autosomal recessive**
- Both alleles of the beta-globin gene must have the mutation

Sickle cell trait

- Genotype AS
- The alleles show **co-dominance**
- Both forms of the protein are observed on electrophoresis
- Both forms of the protein are equally expressed

Hemoglobin electrophoresis shows both the normal Hb (HbA), and sickle Hb (HbS) in sickle cell trait

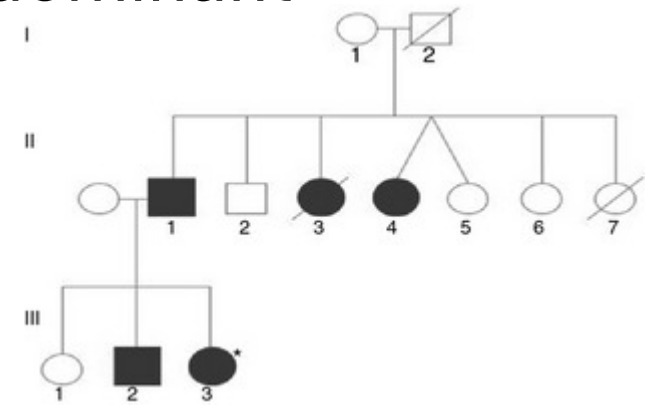


Glu (negative charge) → Val (uncharged)
substitution results in a change in charge of Hb

Pseudo-autosomal dominant

Pseudo-dominance of autosomal recessive disorder

- An autosomal recessive disorder present in individuals in two or more generations of a family, thereby appearing to follow a dominant inheritance pattern.
- An affected parent may have an affected child
 - appears to be **vertical transmission**
- Common explanations include:
 1. High carrier frequency of the disorder
 - **Sickle cell disease in Sub-Saharan areas of Africa**
 2. Higher incidence of consanguinity
 3. Increased carrier frequency because of small population
 - **Geographical, social or religious isolation**

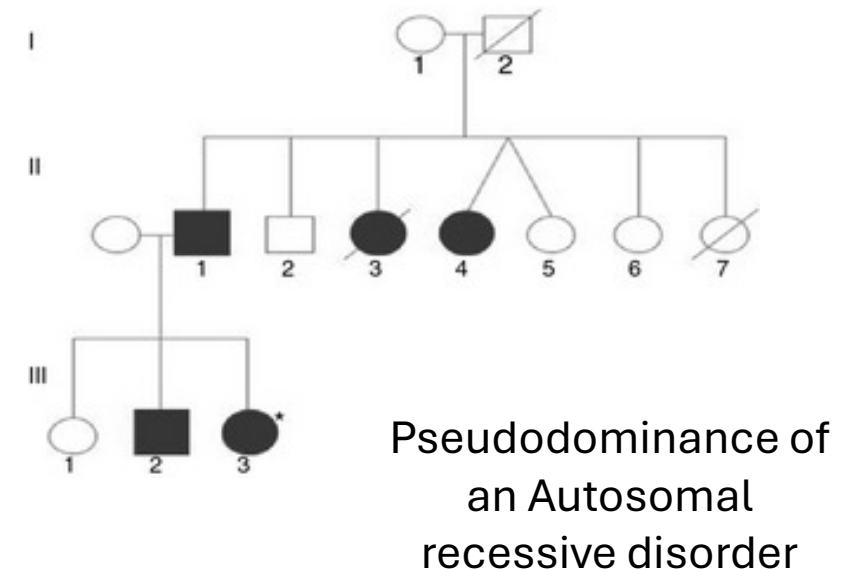
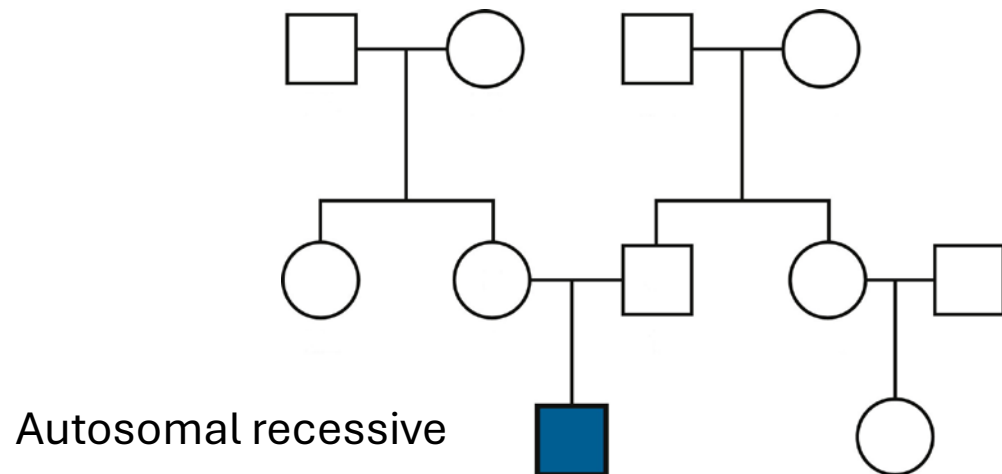


Pseudodominant Inheritance

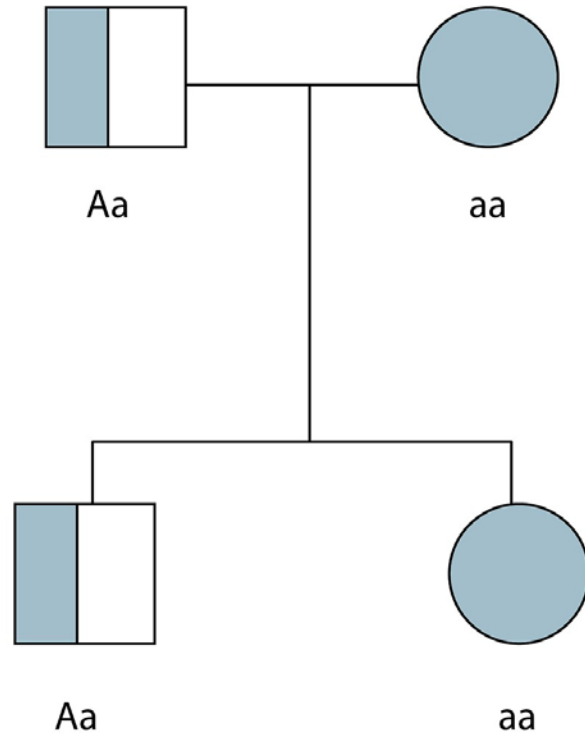
When a carrier (Aa) and an affected individual (aa) mate then there is a $\frac{1}{2}$ risk that the offspring will be affected and a $\frac{1}{2}$ risk to be a carrier.

Called **pseudo-dominance** because it is easy to mistakenly interpret as dominant trait from a simple pedigree (the $\frac{1}{2}$ disease transmission risk suggest dominance at first glance).

- **Pseudo-dominant:** Apparent autosomal dominant inheritance when a heterozygote & homozygote for an autosomal recessive disorder have affected children



Pseudodominant inheritance: **Affected X Carrier**



		Affected parent	
		a	a
Carrier	A	Aa	Aa
	a	aa	aa

Calculate the recurrence risk of the disorder in offspring of I-1 and I-2. Indicate the risk of them having affected children, carriers and normal children (Draw the Punnet square)

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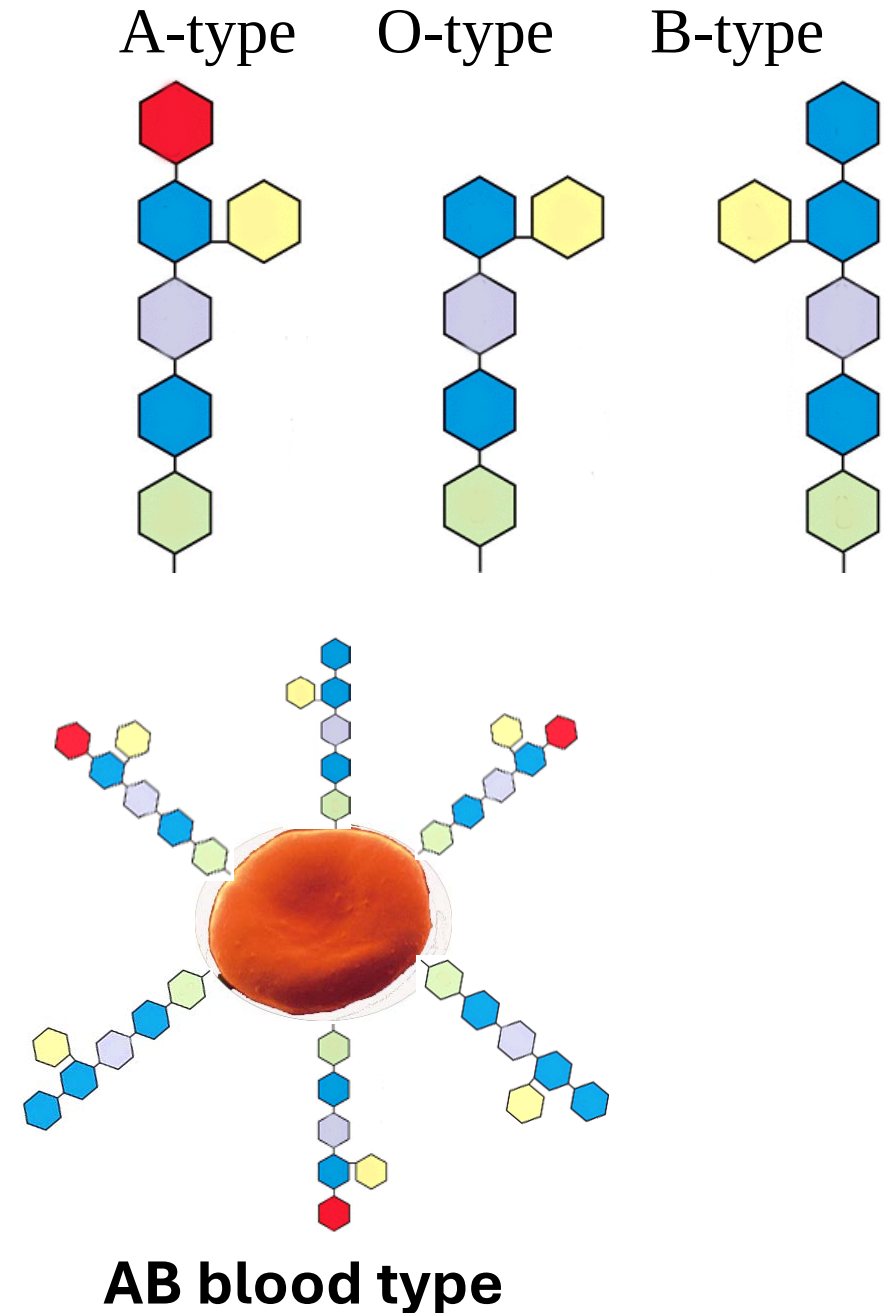
Figure 3.18 Pseudodominant inheritance. A heterozygous male and homozygous female transmit a recessive disorder to their daughter, giving the appearance of dominant transmission. This is most likely to occur when the mutant allele is common in the population.

Factors that increase incidence of an autosomal recessive trait (Pseudo-dominance)

- Consanguinity
- Heterozygote advantage
 1. **Sickle cell disease**
 2. **Cystic fibrosis**
 3. **Hemochromatosis**
- Genetic isolation
 - Geographic, such as outports in Newfoundland
 - Religious, such as Amish, Hutterites, etc.
 - Cultural (language), such as Asian populations in UK
- Assortative mating
 - People associate and marry with “like” (e.g Hearing or visually impaired individuals)

Codominance: Blood group types

- The addition of the sugar is accomplished by specific enzymes
- Enzymes produced by the A and B alleles add a different terminal sugar (O is a null)
- In **AB heterozygotes**, both sugars are present on cell surface at equivalent level
- Both alleles are **equally expressed** on RBCs



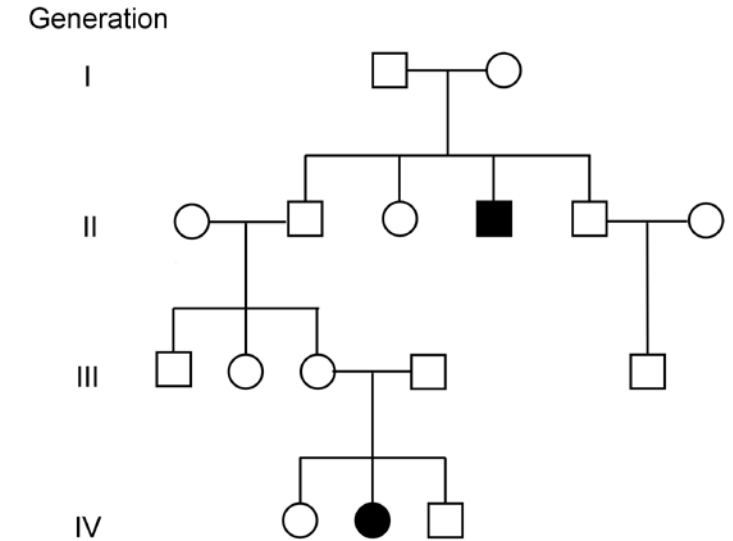
Incomplete dominance (traits)

- Phenotype is intermediate between the parents
- Phenotype is intermediate between the two dominant alleles
 - Father has curly hair
 - Mother has straight hair
 - Baby has wavy hair
- Flowers
 - Red flower x White flower = next generation with Pink flowers



Summary

- Autosomal recessive inheritance
 - Identification of carriers in family
 - Carrier risk calculation
 - Disorders and clinical features
 - Allelic heterogeneity and compound heterozygote
- Pseudodominance of an autosomal recessive disorder



Please email
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you have any questions

Autosomal Recessive disorders

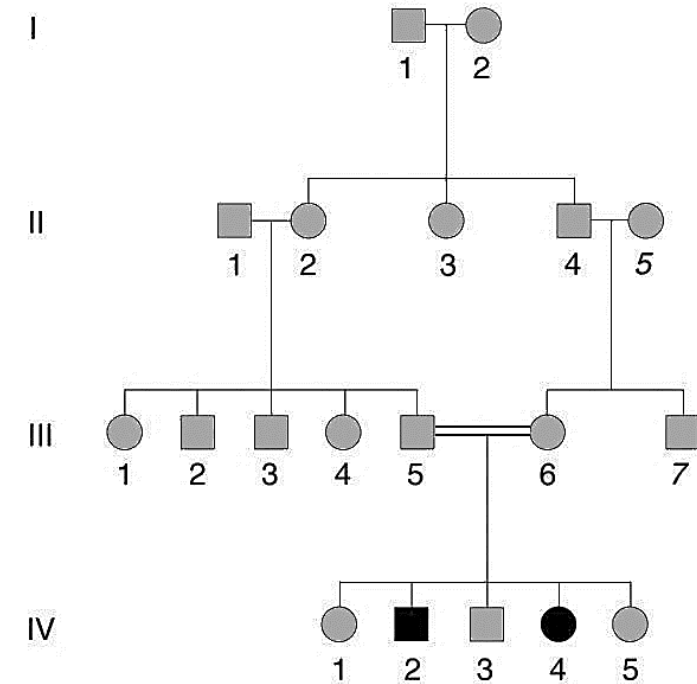
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The family shown by the pedigree has two members with sickle cell disease. What is the probability of IV-3, (36-years-old) being homozygous for the normal form of the gene?

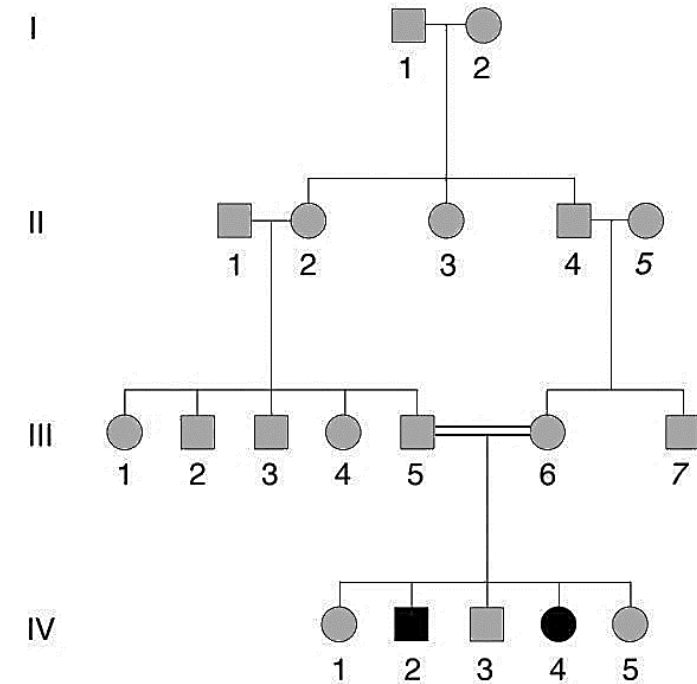
- A. 100%
- B. 75%
- C. 66%
- D. 33%
- E. 25%



The family shown by the pedigree has two members with sickle cell disease. What is the probability of IV-3, (36-years-old) being homozygous for the normal form of the gene?

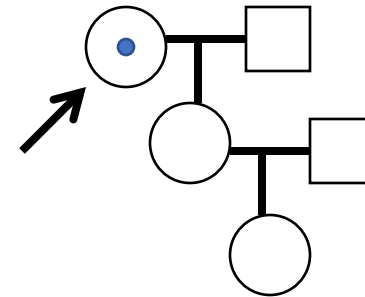
- A. 100%
- B. 75%
- C. 66%
- 😊 D. 33%
- E. 25%

Explanation: We assume here that IV-3 is not aa (homozygous affected); 1/3 chance that he may be homozygous normal (AA); 2/3 risk that he may be a carrier (Aa)



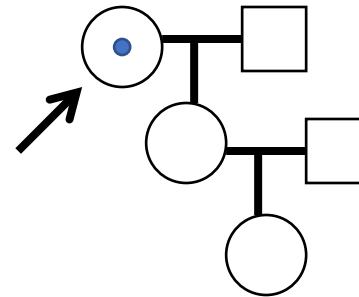
The consultand is a known carrier of a mutant allele of the *ADA* gene. This is known because she had a child with SCID from a different partner, and the causative alleles were identified. What is probability of her child inheriting that same mutant allele?

- A. 100% 0%
- B. 67% 0%
- C. 50% 0%
- D. 33% 0%
- E. 25% 0%
- F. Nil 0%



The consultand is a known carrier of a mutant allele of the *ADA* gene. This is known because she had a child with SCID from a different partner, and the causative alleles were identified. What is probability of her daughter inheriting that same mutant allele?

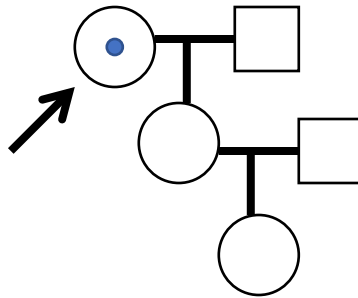
- A. 100% 17%
- B. 67% 17%
- 😊 C. 50% 17%
- D. 33% 17%
- E. 25% 17%
- F. Nil 17%



	A	a
A	AA	Aa
A	AA	Aa

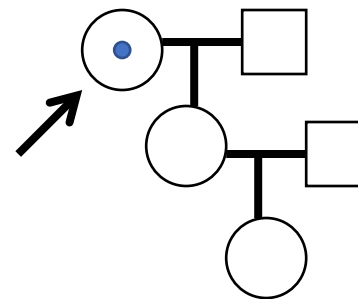
The consultand is identified to be a carrier of the mutant allele of the ADA gene. What is probability of her grandchildren inheriting that same mutant allele?

- A. 100% 17%
- B. 67% 17%
- C. 50% 17%
- D. 33% 17%
- E. 25% 17%
- F. Nil 17%



The consultand is identified to be a carrier of the mutant allele of the ADA gene. What is probability of her grandchildren inheriting that same mutant allele?

- A. 100% 17%
- B. 67% 17%
- C. 50% 17%
- D. 33% 17%
- 😊 E. 25% 17%
- F. Nil 17%



	A	a
A	AA	Aa
A	AA	Aa

$$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$